Linyi (Tiger) Hou

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Education

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign — Jun 2021 - present

B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign — Aug 2017 - May 2021

Work Experience

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483) — Jun 2021 - May 2024

- Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern — May 2019 - Aug 2019

- Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- Automated builds in TortoiseSVN using Python, reducing need for manual integration.

Research

Astrodynamics and Planetary Exploration Group (APEX) — Advisor: Dr. Siegfried Eggl

Interstellar Navigation and Star Identification — Sep 2024 - present

- Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking — Jul 2024 - Mar 2025

- Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars — Nov 2022 - Mar 2025

- Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Aerospace Mission Analysis Laboratory Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV) Jun 2021 - Dec 2023

- Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package PINT to within 50 µs.
- Demonstrated the computational feasibility of (XNAV) in a lost in space scenario.

Velocity-based Initial Orbit Determination Aug 2019 - Apr 2024

- Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- Assessed the performance of velocity IOD methods using Monte Carlo simulation.

Space Systems Optimization Laboratory Advisor: Dr. Koki Ho

Architecture Trade Studies on ISRU Technologies for Human Space Exploration Sep 2018 - Sep 2019

- Worked with two graduate students to discuss the development of a mixed-integer linear programming framework for space logistics optimization.
- Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

Skills

Programming & Typesetting	Python, C/C++, MATLAB/Simulink, Javascript, LaTeX
Libraries & Frameworks	astropy | scipy | pybind11, Eigen | OpenMP, NodeJS
Tools & Databases	Git, GitHub, Slurm, MongoDB, ADQL

Honors and Awards

Aerospace Engineering Alumni Advisory Board Fellowship 2024
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference 2024
American Astronautical Society (AAS) Rocky Mountain Section

Yee Fellowship 2021, 2024
Grainger College of Engineering, University of Illinois Urbana-Champaign

David and Catherine Thompson Space Technology Scholarship 2020
American Institute of Aeronautics and Astronautics (AIAA)

H.S. Stillwell Problem Solving Award 2020
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Robert W. McCloy Memorial Award 2019
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

James Scholar 2018 - 2021
University of Illinois Urbana-Champaign

Journal Articles

[J.5] **L. Hou** et al. “Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars”. In: *The Journal of the Astronautical Sciences* 72.9 (2025). DOI: 10.1007/s40295-025-00485-8.

[J.4] **L. Hou** and Zachary R Putnam. “Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements”. In: *The Journal of the Astronautical Sciences* 71.24 (2024). DOI: 10.1007/s40295-024-00442-x.

[J.3] **L. Hou** and Zachary R Putnam. “A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV”. In: *The Journal of the Astronautical Sciences* 71.6 (2024). DOI: 10.1007/s40295-023-00425-4.

[J.2] Hao Chen et al. “Multifidelity Space Mission Planning and Infrastructure Design Framework for Space Resource Logistics”. In: *Journal of Spacecraft and Rockets* 58.2 (2021), pp. 538–551. DOI: 10.2514/1.A34666.

[J.1] Hao Chen et al. “Integrated In-Situ Resource Utilization System Design and Logistics for Mars Exploration”. In: *Acta Astronautica* 170 (2020), pp. 80–92. DOI: 10.1016/j.actaastro.2020.01.031.

Conference Proceedings

[C.8] **L. Hou** and Siegfried Eggl. “Autonomous Optical Navigation using Astrometry and Photometry”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 25-248. Kaua’i, HI, Jan. 2025.

[C.7] **L. Hou** et al. “Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars”. In: *46th Rocky Mountain AAS GN&C Conference*. AAS-24-012. Breckenridge, CO, Feb. 2024.

[C.6] Alec Auster et al. “Mars Ice Thermal Harvesting Rig & ISRU Laboratory (MITHRIL)”. In: *ASCEND 2022*. Aug. 2022, p. 4249. DOI: 10.2514/6.2022-4249.

[C.5] **L. Hou** and Zachary R. Putnam. “A Norm-minimization Method for Solving the Cold-Start Problem with XNAV”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-560. Charlotte, NC, Aug. 2022.

[C.4] **L. Hou** and Zachary R. Putnam. “Initial Orbit Determination from Sequential Line-of-Sight Velocity Measurements”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-561. Charlotte, NC, Aug. 2022.

[C.3] **L. Hou**, Kevin Lohan, and Zachary R Putnam. “Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 21-280. Virtual, Feb. 2021.

[C.2] Hao Chen et al. “Multi-Fidelity Space Mission Planning and Space Infrastructure Design Framework for Space Resource Logistics”. In: *AIAA Propulsion and Energy 2019 Forum*. 2019, p. 4134. DOI: 10.2514/6.2019-4134.

[C.1] H Chen et al. “Integrated Analysis Framework for Space Propellant Logistics: Production, Storage, and Transportation”. In: *Lunar ISRU 2019-Developing a New Space Economy Through Lunar Resources and Their Utilization* 2152 (2019), p. 5003. URL: https://www.hou.usra.edu/meetings/lunarisru2019/pdf/5003.pdf.